

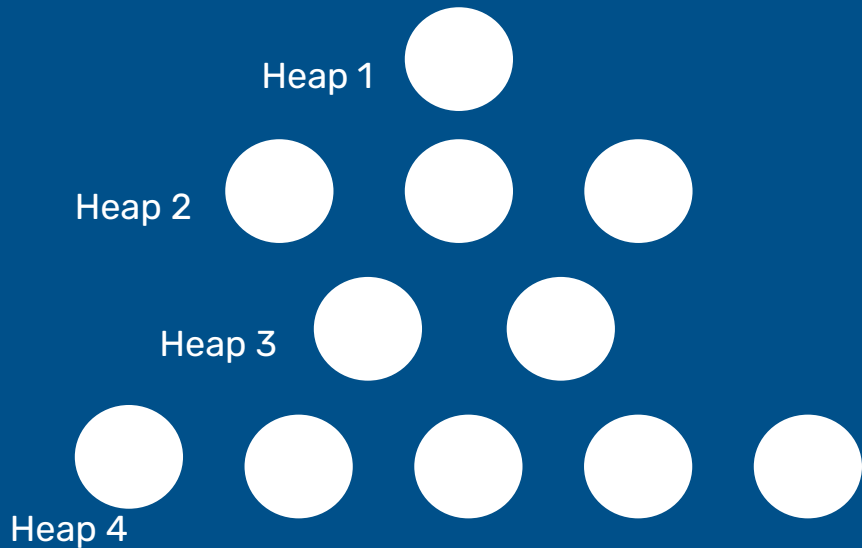
Nim Game Project

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Quick Review

What is Nim?

- **Each player takes as many object from distinct heaps**
- **Normal Play**
 - Person who takes last object wins
- **Misere**
 - Person who takes last object loses



Mathematical Significance

Nim Game

- Normal Play and Misere Similarities
 - Calculate nim-sum
 - If equal to 0, losing position
 - If not equal to 0, finish every move with nim-sum of 0

Mathematical Significance

Nim Game

[3 1 4]

==

Nim-Sum = $011 \text{ xor } 001 \text{ xor } 100$

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Nim-Sum = 110

==

$011 \text{ xor } 110 = 101 = 5$

$001 \text{ xor } 110 = 111 = 7$

$100 \text{ xor } 110 = 010 = 2$

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Since only $2 < 4 \rightarrow [3 \ 1 \ 2]$

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New Nim-Sum = $011 \text{ xor } 001 \text{ xor } 010 = 0$

Different Versions

Normal Play and Misere Differences

Normal Play

- When reach 1 heap or 2 heaps optimal strategy changes
- **1 heap** = take all
- **2 heap** = if heaps different sizes, take whichever has more to match the other
 - If not different sizes, in losing position

Misere

- **1 heap** = reduce to size 1
 - If already size 1, you lose
- **2 heap and either is of size 1** = take the entire heap not of size 1 (or 1 if both are)
- **In endgame (all heaps size 1 except 1)** = reduce to make odd total

What did I do?

Project!

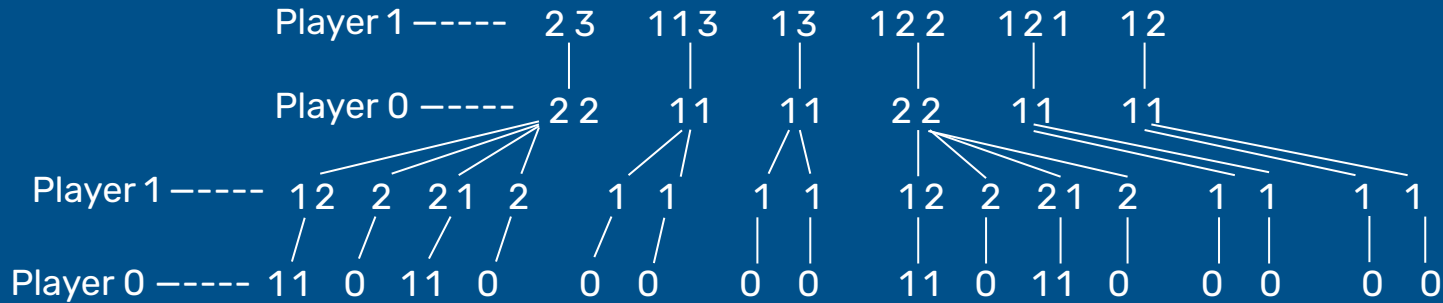
- Create 2 programs, one for normal play and one for misere
- Gave computer optimal solution
- When other player's turn, ran all possible moves a player could make
- Outputted entire list of results
- **In ACL2s!**

DEMO!

Tree Mapping Of What Program Doing

Normal Play

Normal Play - start with Player 1, 1 2 3, -1



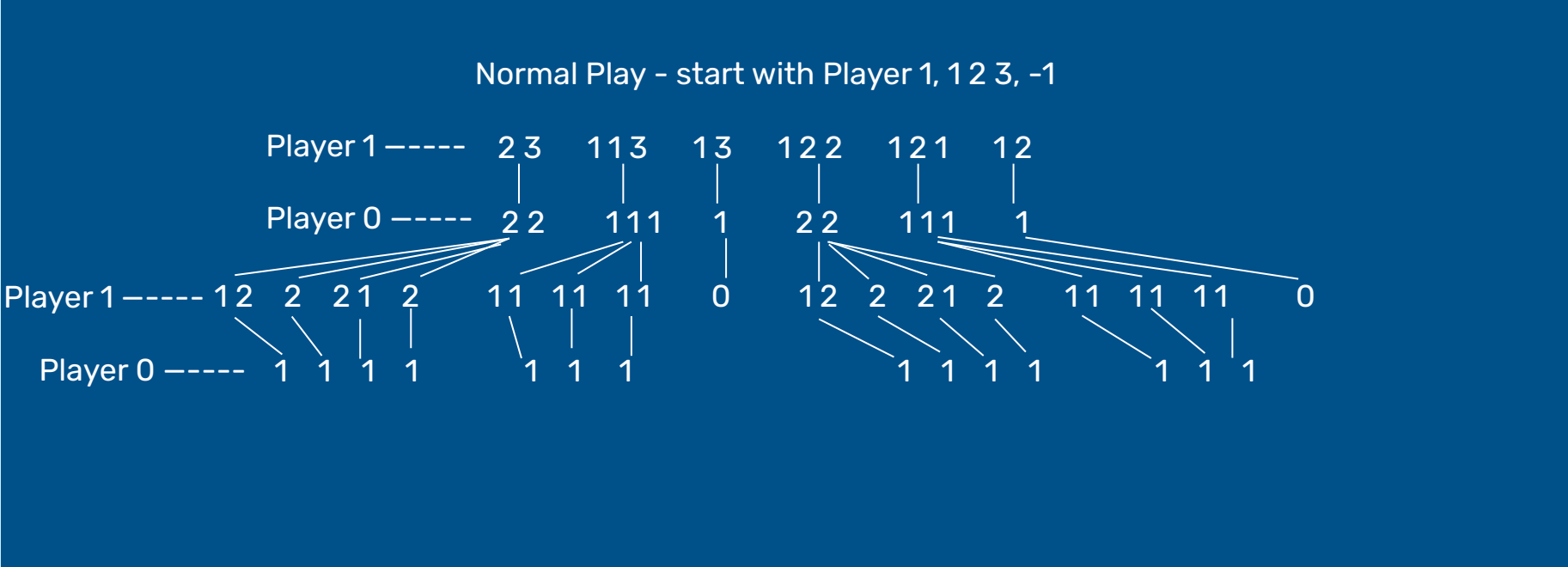
Actual Output

Normal Play Output

[illegible]

Tree Mapping Of What Program Doing

Misere

[illegible]

Actual Output

Misere Output

```
ACL2S !>VALUE (play-game '(1 (1 2 3) -1))
((0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0))
ACL2S !>VALUE (play-game '(0 (1 2 3) -1))
((0 nil 0)
 (0 nil 0)
 (1 nil 1)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0)
 (0 nil 0))
```

Any Questions?